GOBI IF1200 Fieldbook

APPN Aerial SOP — Locked revision 1.03

APPN – GOBI IF1200 Fieldbook

For a high-level description of the GOBI sensor package and how it sits alongside the other APPN platforms, see the Platforms Overview.

For the GOBI fieldbook covering the DJI M350, see GOBI M350 Fieldbook.

This fieldbook provides a standardised operational guide for APPN GOBI UAV deployments **on the Inspired Flight IF1200A**, supporting safe flight operations, consistent sensor configuration, and high-integrity data capture. It is intended for trained APPN staff conducting hyperspectral, LiDAR, RGB, and GNSS-INS data acquisitions, and promotes repeatability, transparency, and confidence in downstream data analysis across APPN operations.

> **Important**
>
> **This protocol must be followed for all standard APPN GOBI IF1200 UAV flights.** Adherence to these procedures is essential to ensure operational safety, data integrity, and comparability of datasets across deployments. For any flights that **fall outside standard operating procedures**, detailed records must be kept documenting all deviations, including the specific settings changed, the rationale for those changes, and any anticipated implications for data quality or analysis.

Document Structure

1. Equipment Checklists — what to bring to the field.
2. Flight Planning – Inspired Flight IF1200A — survey / capture polygons and QGroundControl flight planning.
3. Pre-Flight — weather and airspace checks, panel and GCP layout, aircraft setup.
4. GOBI Sensor Configuration — sensor power-up, exposure, dark reference, and mission start.
5. Flight Operations — takeoff, dynamic alignment, autonomous mission, and landing.
6. Post-Flight Sensor Configuration — stopping the mission and confirming data on the sensor.
7. Data Confirmation — expected file counts and sizes by sensor.
8. Post-Flight — pack-down of aircraft, payload, and ground reference kit.
9. Offload Data — downloading from the sensor and storing under the APPN folder structure:

- Data storage, processing & validation

10. Appendix — FTP & service reference, Headwall polygon tool workflow, static IP setup, and IP address reference.

Equipment Checklists

> **Note**
>
> Ensure batteries for all equipment are fully charged before heading to the field. Ensure charging cables are available for necessary equipment.

Warning

Cables and adaptors (particularly ethernet and USB) are known to fail in the field on hot days. Bring spares of all critical cables and adaptors wherever possible.

- ☐ **Aircraft (Inspired Flight IF1200A)**
- ☐ Aircraft batteries
- ☐ Landing gear
- ☐ Landing pad
- ☐ Spare parts
- ☐ Tools
- ☐ Logbook
- ☐ **Radio Control Transmitter / Ground Control Station**
- ☐ **GRYFN Gobi**
- ☐ Gobi system
- ☐ Power cables, chargers
- ☐ Ethernet cable
- ☐ SD card
- ☐ *Capture* polygon files
- ☐ **Ground reference kit**
- ☐ 2 x Reflectance calibration panels (11%, 30%, 56%, 82%)
- ☐ 1 x Calibration validation panels (20%, 45%)
- ☐ 5 × Propeller Aeropoints or other ground control points (GCPs) and/or RTK GNSS system
- ☐ 2 × folding tables to elevate panels
- ☐ *If over 50 km from CORS base station*, a portable RTK base station (link to GRYFN gitbook)
- ☐ **Accessories**
- ☐ Safety gear (signage and high-vis vests)
- ☐ Aeronautical radio
- ☐ Field laptop and spare batteries
- ☐ External storage media
- ☐ Water, food, esky, sunscreen, bug spray, first aid kit, etc.
- ☐ Spirit bubble, spirit level (or angle measurement) and measuring tape
- ☐ Portable fan (for cooling GOBI during data offload)
- ☐ External power brick (for charging UAV RC)
- ☐ Anemometer (e.g. Kestrel 2000 Wind Meter)

Note

The **Ground reference kit** above assumes the Standard Dual ELM panel flight design. Other flight designs may require additional equipment (e.g. extra reflectance panel sets, additional GCPs and folding tables for multi-flight captures, or extra targets for validation flights). See the Standard Flight Procedure and the Validation Flight Procedure for the full equipment requirements of each design.

Flight Planning – Inspired Flight IF1200A

Important

Ensure that you apply for UAV flight approvals for locations and dates of flights well in advance. Always use the **latest version** of the GRYFN Flight Calculator when planning flights.

1. Using a GPS survey system (Emlid, Trimble…) or a GIS software, create a polygon of the area of interest. Make sure your polygon includes the areas where you will place your calibration panels and GCPs, with an additional 5 m buffer to avoid incomplete data.
2. Save this polygon twice as a KML — once as a *survey* polygon and once as a *capture* polygon.

Note

Buffer the *capture* polygon **perpendicular to the flight direction** by at least one flight-line spacing on each side. When a flight line runs immediately adjacent to the capture-polygon boundary, the aircraft can drift out of the capture region and hyperspectral triggering stops, losing data along that edge. A perpendicular buffer provides margin for trajectory variation and ensures full capture on the outermost lines. **Increase the buffer further when flight lines cross the polygon boundary at an angle** (e.g. north–south APEx flights): the aircraft enters and exits the polygon at an angle, so a triggered frame can fall half inside and half outside the capture region.

3. If using QGIS, export the polygon as a KML (in Geometry, select *include z-dimension*, and ensure the CRS is set to WGS 84). Import the *capture* KML into the HPI Polygon Tool and export. This polygon sets the activation of the hyperspectral sensor within Hyperspec3.
4. Import the *survey* polygon into QGroundControl.
5. Using both QGroundControl and the GRYFN flight calculator, determine the speed, altitude, and frame period required to survey the area of interest. See the Standard Mission Parameters table below for examples of common flight types.

- Do not go below 2 m/s or above 8 m/s for stability reasons (GRYFN). Speeds greater than 8 m/s lead to excessive aircraft pitch and speeds less than 2 m/s accentuate the impacts of wind on aircraft stability and cause a visibly less smooth trajectory.
- Altitude and speed will be tested and recommended from APEx results.
- Ensure the frame period is at a minimum of 20% oversampling. Frame period sets the along-track sampling and is separate from side overlap.
- Ensure **side overlap is driven by the VNIR requirement, not the RGB camera** (see the important note under Standard Mission Parameters). For hyperspectral missions (Types 1–3), plan at **80% RGB side overlap**, which yields approximately **37% VNIR side overlap**; the flight-line spacings in the table below are derived from this requirement.
- Ensure the *turnaround distance* is 2× flight speed ($> 3\times$ at > 6 m/s).

6. Ensure flight lines are in the direction of planting (GRYFN).

Standard Mission Parameters

Standard Mission Type	Scenario	Altitude (m)	Speed (m/s)	Frame Period (Hz)	Distance between flight lines — IF1200 (m)
Type 1	Plant counting / small structures (intra-plot differences)	30	2.1	5.32 (20% oversampling)	9
Type 2	Plant breeding experiments (inter-plot differences)	50	3.2	5.09 (30% oversampling)	15
Type 3	Large landscape measurements (strip trials, hyperspectral transects)	80	5.1	5.11 (30% oversampling)	24
Type 4	Surveys using LiDAR and RGB only (ecosystem measurements, forestry)	100	9	N/A	37

Important

Overlap must be driven by the VNIR (hyperspectral) data-quality requirement — RGB overlap alone is not a sufficient measure of acceptable sidelap. The flight-line distances above are derived from **80% RGB side overlap** (via the GRYFN flight calculator), which resolves to approximately **37% VNIR side overlap**. Historically the 75% RGB setting was assumed to guarantee adequate VNIR sidelap; in practice it produces *very low* VNIR overlap and has caused missing pixels, linear data holes, and stronger edge/directional effects in recent flights.

VNIR side-overlap targets:

- ~**37%** — delivered by the spacings above; treat as the practical minimum.
- $\geq$ **40%** — preferred, for robustness under wind and trajectory variation.
- $\geq$ **50%** — every ground point is imaged by **at least two flight lines**, greatly increasing redundancy and resilience to trajectory error. Reduce line spacing further where this level of redundancy is required.

RGB frame cameras remain the limiting factor for the orthomosaic — GRYFN recommends an absolute minimum of 70% overlap/sidelap, with 80% giving better feature matching. These spacings bring the GOBI IF1200 into alignment with the GOBI M350 recommendations. **RGB-only missions (Type 4)** have no VNIR requirement.

Note

Frame-period values may differ slightly between the **GRYFN Flight Calculator** and the **Gryfn WebUI**. The WebUI uses each sensor's *calibrated* focal length, whereas the calculator uses the manufacturer's nominal value (VNIR ~12.6 mm, RGB ~24 mm). Calibrated focal lengths are typically slightly shorter, marginally reducing the computed frame rate and overlap, but the difference is negligible (~0.01 ms in frame period, ~0.02% in sidelap) and is absorbed by the oversampling and sidelap margins. Either source is acceptable; the values in the table above are from the flight calculator.

Note

Type 1 (30 m) uses 20% oversampling, not the 30% used for Types 2–3. At 30 m the finer ground sampling requires a higher frame rate, so 30% oversampling pushes the VNIR frame rate close to the sensor's maximum, which GRYFN does not recommend. 20% oversampling keeps a safe margin below the maximum frame rate while still meeting the minimum oversampling requirement.

Note

The IF1200 flies **without RTK**, so positional error can be around ± 1 m. At low altitude this materially erodes effective overlap: e.g. a 30 m flight planned at 9 m line spacing can, if two adjacent lines drift ~1 m apart, open to ~10 m and fall to very low VNIR sidelap. Outright gaps are unlikely but overlap can become marginal. Higher altitudes have proportionally more spacing buffer and are less sensitive; favour extra overlap (or tighter spacing) for low-altitude non-RTK flights.

Pre-Flight

1. Conduct airspace and weather checks.

- No cloud cover.
- Maximum wind speed for quality hyperspectral capture is **6 m/s (~22 km/h)**. Any wind speed over **5 m/s (~18 km/h)** should be recorded.
- Do not operate in conditions below 0 °C or above 40 °C.
- Try to ensure the sun is within ±20° of solar noon (approximately 2 hours before or after noon). **However**, *this depends on time of year and latitude — check NOAA solar calculator if unsure.*

2. Turn on the aeronautical radio and set to local CTAF (find in ERSA).
3. *(Optional)* Set up the Emlid RTK (install a peg or permanent GCP below the base station for recurring flights), let it run for at least 15 minutes, and start recording the RINEX file before flying.
4. Set up reflectance panels, the validation panel, and GCPs following the Standard Flight Procedure. In summary:

- Use the Dual ELM panel flight layout by default (mandatory for missions > 15 minutes — noting that IF1200 missions are typically < 30 minutes, so a calibration panel is overflown approximately every 15 minutes).
- Fall back to the Single ELM panel flight layout only when a second panel set is unavailable.
- Deploy the 2-panel validation set with its paired elevated + ground GCPs per Validation Panel Setup.
- Follow the Panel Setup rules (level folding table, panels in brightness order aligned with the flight direction, record panel-surface height).
- Place 5 GCPs per Ground Control Points — in alternate flight lines, next to the calibration panels.
- When flying multiple missions, panels and GCPs must be present in every flight (panels overflown approximately every 15 minutes).

Figure: Correctly deployed 4-panel ELM reflectance set on a levelled folding table. See the Standard Flight Procedure for the full panel and GCP layout diagrams.

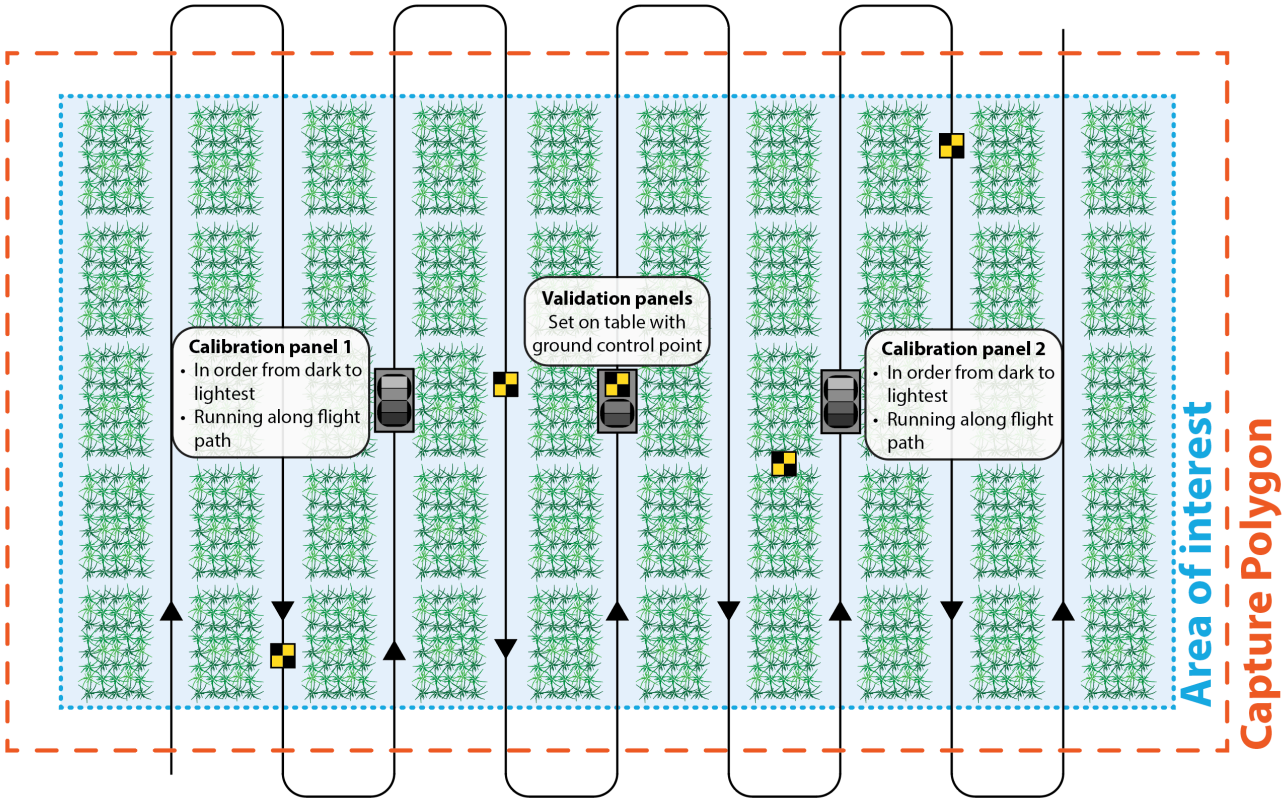

Figure: Dual ELM panel flight layout — default layout for GOBI IF1200 surveys. See the Dual ELM panel flight section of the Standard Flight Procedure for the full rationale and alternative layouts.

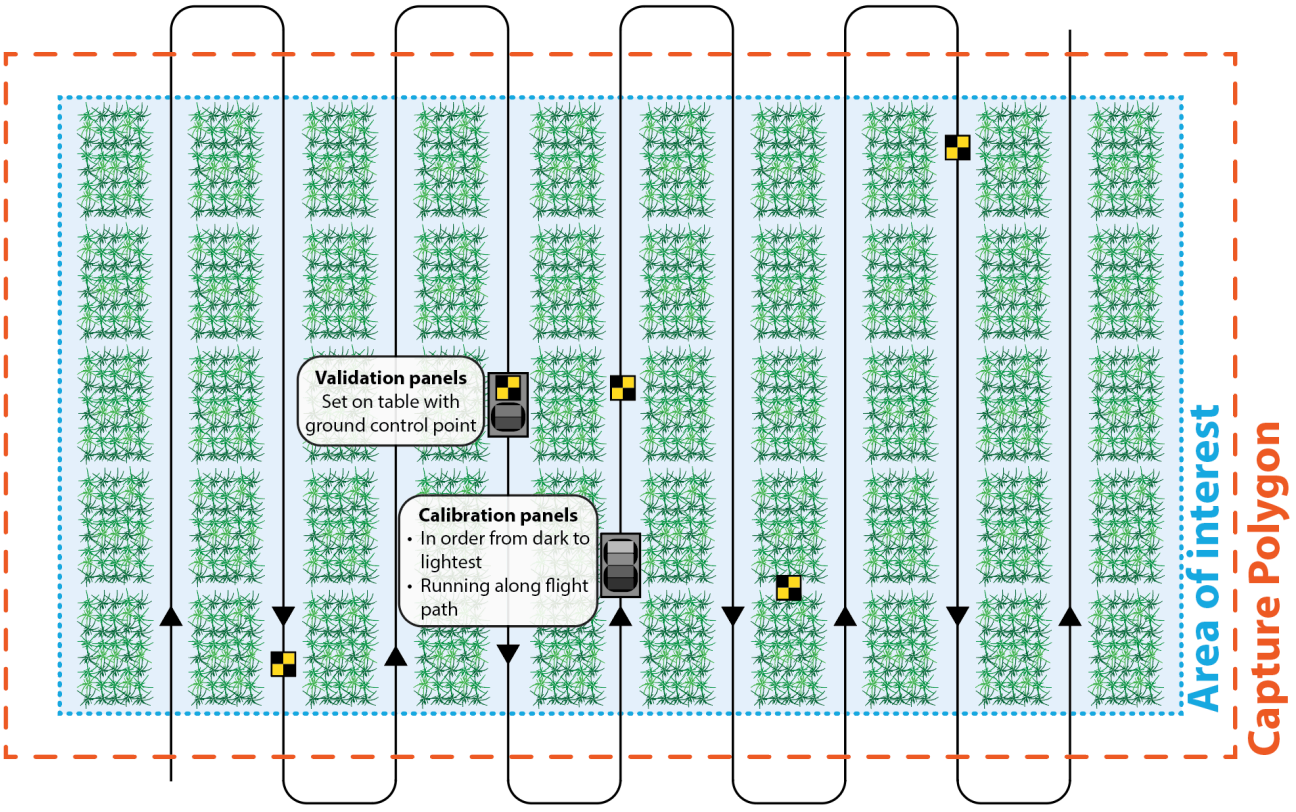

Figure: Single ELM panel flight layout — fallback when a second panel set is unavailable. See the Single ELM panel flight section of the Standard Flight Procedure for full guidance.

6. Set up the landing pad and UAV in a safe RTH location.

> **Important**
>
> In dusty environments, an additional tarp must be used under the landing pad.

7. On the controller, set up a safe UAV RTH location, RTH altitude, and other geo-fencing settings on the IF1200.
8. Attach payload to the aircraft:

> **Caution**
>
> The IF1200 dovetail has no hot-swap protection, so ensure the IF1200 is powered off when attaching or removing the sensor (more details).

- Check that all lenses and sensors are clean. If not, use professional lens cleaning wipes (e.g. Zeiss Lens Wipes, recommended by GRYFN) to clean them.
- Connect the power cable from aircraft to payload (non-standard payload bus only).
- Connect both GNSS antenna cables to the correct ports (match A1 and A2).
- Remove RGB and Nano HP lens caps.
- Insert RGB SD card.

9. Power on the radio controller; check battery status.
10. Launch QGroundControl on the IF1200 controller.
11. Review the flight plan, checking operational height and double-checking that area of interest, GCPs, and reflectance panels are all within the capture polygon.
12. Power on the aircraft; confirm connection to GCS and battery status; ensure Remote ID is enabled.
13. The IF1200 does not use RTK; ensure a minimum of 8 satellites and GPS lock before flying.
14. Upload the flight plan to the aircraft.

GOBI Sensor Configuration

1. Place the 80% exposure reference panel under the Nano HP lens; avoid casting shadows.

- Place the drone legs upon exposure-angle cones, ensuring a fixed angle is achieved each exposure setting. **(angle provided through SIF excellence)**
- The maximum distance from the VNIR HS sensor to the GOBI reflectance panel is 40 cm. This is calculated based on the FOV and a 20 cm panel size.

2. Connect to GOBI Wi-Fi or connect the ethernet cable to the payload. *If using ethernet, ensure a static IP address for the sensor at* `10.0.65.2` (or anything other than `50`, `100`, and `128`, and must be less than `255`).
3. Navigate to the GOBI WebUI at `10.0.65.50`.
4. Ensure a valid altitude is shown in the top-right corner of the WebUI (~10–20 m error is acceptable). (accurate position/elevation should be within ~30-60 seconds from power on)
5. Name the mission, using the convention `YYYYMMDD_XXXX` (`YYYY` = year; `MM` = month; `DD` = day, must be 2 digits; `XXXX` = a short reference or abbreviation for the job).
6. Ensure the correct UAV (IF1200) is selected in the dropdown box. **This is very important.**

> **Caution**
>
> Selecting the incorrect drone will render all data captured unrecoverable

7. Scroll down to polygon settings. Ensure triggering is set by **Polygon** in the dropdown menu. Set min and, optionally, max altitude triggers (recommended to set a minimum of at least 20 m to prevent unnecessary data capture).

8. Import the polygon KML or select from previously flown missions using the dropdown menu. Inspect the map to ensure it aligns with expectations.
9. Return to the top of the WebUI, open the Flight Calculator, and update flying height / flight speed; adjust the oversampling buffer (must be at least 20%, recommended setting is 30%). **(value may change through hyperspectral excellence)** Frame period will be automatically updated.
10. Adjust exposure until ~95% spectral saturation at the **peak of the 90th percentile line** without exceeding the frame period. Use the lowest gain mode that still provides sufficient saturation. **(value may change through hyperspectral excellence)**
11. Place the lens cap on the Nano HP.
12. Click **Collect Dark Reference**; wait to finish.
13. **Remove the lens cap.**
14. Remove the exposure reference panel and return it to its case.
15. Press **Start Mission**.
16. Remove the ethernet cable (if using) and begin flight operations.

Flight Operations

1. Ensure the aircraft is in **Position** flight mode.
2. Sync / verify the flight plan in QGroundControl; double-check the flight plan.
3. Clear people/objects away from the UAV.
4. Notify crew/observers that takeoff is beginning.
5. Begin manual takeoff, check stick controls work, and then fly to ~12 m AGL; **do not exceed** the trigger altitude (20 m default) before mission start.
6. Perform dynamic alignment (one to two figure-8 patterns) at a recommended speed of **5 m/s**, at a lower height than capture height and outside of the capture polygon area.
7. Enable autonomous mission.
8. Monitor UAV battery voltage/percentage in flight.
9. After mission, switch back to **Position** mode to regain manual control.
10. Lower to capture altitude and perform post-mission dynamic alignment (figure-8).
11. Land and disarm UAV.
12. Leave the UAV, transmitter, and sensor powered on; begin post-flight checks. Ensure data transfer is finished before turning off the sensor.
13. **Do not hot-swap the batteries.** Treat each flight as a new flight.

Post-Flight Sensor Configuration

1. Reconnect to GOBI Wi-Fi or ethernet cable.
2. Open a browser to `10.0.65.50` or refresh the UI.
3. Press **Stop Mission**.
4. Replace the Nano HP and RGB lens caps.
5. In the WebUI, ensure all data has captured correctly (see the Data Confirmation table below).

Data Confirmation

Sensor type	File count	File size	Notes
GNSS-INS	1 file	~1.5 MB per minute	A new file will be created if time rolls over the hour (UTC time, not capture time).
LiDAR	1 per minute	~175 MB per file	First & last file may be smaller. Files may be smaller if over a low-reflectivity object.
VNIR	Mission time (s) / Frame Period value ~ number of data cubes	Several GB even for short flights.	Check that `frameindex`, `imu_gps`, and `settings` files all exist.
RGB	One image every 2 s* (by default)	~30–70 MB per image	Check the event file vs the number of images.

Post-Flight

1. Power off the aircraft, then the controller. Inspect the aircraft for any damage and report for maintenance if necessary.
2. Disconnect the ethernet cable from the payload, if using.
3. Disconnect the aircraft batteries.
4. Disconnect the payload power cable.
5. Disconnect the payload GNSS cables.
6. Undo the mounting clamp, remove the payload, and place it in its case.
7. Pack up the aircraft and all gear from site, double-checking against the checklist at the start of this document to ensure everything is accounted for.

Offload Data

1. Open WinSCP, FileZilla, or similar FTP software.
2. Connect to the GRYFN Gobi via `gryfn@10.0.65.50` (password: `gryfn`).
3. Raw GNSS & LiDAR PCAP location: `/data/{mission name}`.
4. VNIR data location: `/data/capturedData/captured/{dateTime}`.
5. GOBI logs location: `/data/gryfn.log.{date}`.
6. RGB images are stored on the camera SD card.
7. Download GNSS, LiDAR, logs, and RGB after each mission; download hyperspectral data later due to its size.
8. Clear data directories after download and backup to avoid filling the 500 GB SSD.

Data storage, processing & validation

All paths below follow the APPN folder structure. Formal paths use the wiki's placeholder syntax; an example follows each.

1. Downloaded data should be stored in the correct `T0_raw` folder.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/
```

2. Data from the GCP points should be saved in the `T0_raw/Vault` folder (location may change).

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/Vault/
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/Vault/
```

3. Important information and issues should be recorded in the `FieldNotes.txt` file.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
```

{SensorPlatform}/{YYYYMMDD}/FieldNotes.txt

Example:

./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/FieldNotes.txt

4. Per-run information (e.g. APEx test conditions) should be stored in `RunOverview.csv`.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/RunOverview.csv
```

Example:

./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/RunOverview.csv

Important

When data from a failed run is being kept (e.g. debugging with GRYFN), the `RunFailed` boolean column of `RunOverview.csv` must be set to `True`.

5. The bundled `.graw` should be saved in the same `T0_raw` folder.
6. The `.gpro` should be generated using the APPN GPT Pipeline (see the Processing Pipelines document and the `Gobi_standard_pipeline_v1.0.yaml` in the adjacent `yaml/` folder) and stored in the adjacent `T1_proc` folder.

Formal path:

```
./{Node}/
{YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
{YYYYSiteName[_F|C]}/
{SensorPlatform}/{YYYYMMDD}/run_XX/T1_proc/
```

Example:

./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T1_proc/

7. Standard QA process should be performed following this guide.

Note

Missing pixels along VNIR image edges can result from insufficient overlap, trajectory irregularities, overly restrictive KML boundaries, or a restrictive processing extent. Where practical, process with no fixed extent (or against a generously buffered capture polygon) so valid edge data is not clipped.

Appendix

FTP & Service Reference

Service	Protocol	IP	Port	User	Password
GOBI	FTP	`10.0.65.50`	22	`gryfn`	`gryfn`
SBG	FTP	`10.0.65.100`	21	`operator`	*(none)*
Headwall polygon	HTTP	`apps.headwallphotonics.com`	—	—	—

Headwall Polygon Tool — Workflow

1. Import KML.

- Must be in Google Earth format.

2. Export KML.

3. Save KML.
4. In File Explorer, rename the KML as the survey name.

Setting a Static IP Address

If connecting over an ethernet connection, a static IP address will need to be set:

1. Open *Network and Internet Settings*.
2. Open *Change Adapter Options*.
3. Open *Properties* for the Ethernet port/adapter.
4. Open *Properties* for IPv4.
5. Set IP address to `10.0.65.2`.
6. Set subnet mask to `255.255.255.0`.
7. Press **OK**.

IP Address Reference

Service	Address
GRYFN WebUI	`10.0.65.50`
HSInsight	`10.0.65.50:8080`
Ouster	`10.0.65.128`
SBG	`10.0.65.100`